Note that speed is a scalar quantity and velocity is a vector quantity (scalar and vector will be discussed in Unit 2) but both of them have the same dimensional formula.

$$\text{Acceleration}, \vec{a} = \frac{\text{velocity}}{\text{time taken}} \Rightarrow [\vec{a}] = \frac{\text{LT}^{-1}}{\text{T}} = \text{LT}^{-2}$$

Acceleration is velocity per time.

Linear momentum or Momentum,

$$[\vec{p}] = m\vec{v} \Rightarrow [\vec{p}] = MLT^{-1}$$

$$\text{Force}, \vec{F} = m\vec{a} \Rightarrow [\vec{F}] = MLT^{-2} = \frac{\text{Momentum}}{\text{time}}$$

This is true for any kind of force. There are only four types of forces that exist in nature viz strong force, electromagnetic force, weak force and gravitational force. Further, frictional force, centripetal force, centrifugal force, all have the dimension $MLT^{-2}$.

$$\text{Impulse}, \vec{I} = \vec{F}t \Rightarrow [\vec{I}] = MLT^{-1}$$

$$= \text{dimension of momentum}$$

Angular momentum is the moment of linear momentum (discussed in unit 5).

$$\text{Angular Momentum}, \vec{\text{L}} = \vec{r} \times \vec{p} \Rightarrow [\vec{\text{L}}] = ML^2T^{-1}$$

Work done,

$$W = \vec{F} \cdot \vec{d} \Rightarrow [W] = ML^2T^{-2}$$

Kinetic energy

$$KE = \frac{1}{2}mv^2 \Rightarrow [KE] = \left[\frac{1}{2}\right][m][v^2]$$

Since, number $\frac{1}{2}$ is dimensionless, the dimension of kinetic energy $[KE] = [m][v^2] = ML^2T^{-2}$. Similarly, to get the dimension of potential energy, let us consider the gravitational potential energy, $PE = mgh \Rightarrow [PE] = [m][g][h],$, where, $m$ is the mass of the particle, $g$ is the acceleration due to gravity and $h$ is the height from the ground level. Hence, $[PE] = [m][g][h] = ML^2T^{-2}$. Thus, for any kind of energy ( such as for internal energy, total energy etc ), the dimension is

$$[Energy] = ML^2T^{-2}$$

The moment of force is known as torque, $\vec{\tau} = \vec{r} \times \vec{F} \Rightarrow [\vec{\tau}] = ML^2T^{-2}$ (Read the symbol τ as tau – Greek alphabet). Note that the dimension of torque and dimension of energy are identical but they are different physical quantities. Further one of them is a scalar (energy) and another one is a vector (torque). This means that the dimensionally same physical quantities need not be the same physical quantities.

**Note**

**1.** We may come across dimensions in different situations in physics, so we often confuse with the term 'dimension'. For instance, we come across terms like 'dimension of energy', 'motion in one dimension' and 'dimension of atom'. It should be kept in mind that dimension of physical quantity means expressing physical quantity in terms of exponent of the base quantity. Motion in one dimension, two dimensions and three dimensions implies that it gives dimension of space. Dimension of atom implies the size of the atom. So, simply writing dimension is meaningless. Hence, the meaning should be taken with the context we write.

**2.** All the trigonometric functions like sinθ, cosθ etc. are dimensionless (θ is dimensionless), exponential function $e^x$ and logarithm function *ln x* are dimensionless ($x$ must be dimensionless). Suppose we expand a function in series expansion (finite or infinite) which contain terms like, $x^o, x^1, x^2, \ldots$. then $x$ must be dimensionless quantity.



UNIT-1(XI-Physics_Vol-1).indd 30 05-01-2022 22:12:00